FACULTY OF ENGINEERING AND ARCHITECTURE

DETOUR

Technical report

Course: Design Thinking

Students: Team 01

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Short summary

This is a section without section number. Unnumbered sections are ignored in the table of contents (toc) by default. Therefore an additional statement is necessary to make sure that this section is added to the toc.

1 Introduction

Your introduction goes here! Simply start writing your document and use the Recompile button to view the updated PDF preview. Examples of commonly used commands and features are listed below, to help you get started.

Once you're familiar with the editor, you can find various project setting in the Overleaf menu, accessed via the button in the very top left of the editor. To view tutorials, user guides, and further documentation, please visit our help library.

2 Some examples to get started

2.1 How to create Sections, Subsections and Subsubsections

Simply use the section, subsection and subsubsection commands, as in this example document! With Overleaf, all the formatting and numbering is handled automatically according to the template you've chosen. If you're using Rich Text mode, you can also create new section and subsections via the buttons in the editor toolbar.

2.1.1 Example subsubsection

Here is an example of a subsubsection.

2.2 How to include Figures

First you have to upload the image file from your computer using the upload link in the file-tree menu. Then use the includegraphics command to include it in your document. Use the figure environment and the caption command to add a number and a caption to your figure. See the code for Figure 1 in this subsection for an example.

Note that your figure will automatically be placed in the most appropriate place for it, given the surrounding text and taking into account other figures or tables that may be close by. You can find out more about adding images to your documents in this help article on including images on Overleaf.

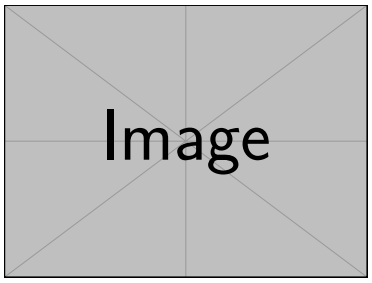


Figure 1: This is an example image.

2.3 How to add Tables

Use the table and tabular environments for basic tables — see Table 1, for example. For more information, please see this help article on tables.

Item	Quantity
Widgets	42
Gadgets	13

Table 1: An example table.

2.4 How to add Lists

You can make lists with automatic numbering …

1. Like this,
2. and like this.

… or bullet points …

- Like this,
- and like this.

2.5 How to write Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

2.6 How to change the spell check settings

To change the spell check language, simply open the Overleaf menu at the top left of the editor window, scroll down to the spell check setting, and adjust accordingly.

2.7 How to add Citations and a References List

You can simply upload a `.bib` file containing your BibTeX entries, created with a tool such as JabRef. Alternatively, you can manually appendix BibTex entries to the `references.bib` file available on e.g. Google Scholar. You can then cite entries from it, like this: **greenwade93** or [**greenwade93**].

2.8 Good luck!

We hope you find Overleaf useful, and do take a look at our help library for more tutorials and user guides! Please also let us know if you have any feedback using the Contact Us link at the bottom of the Overleaf menu — or use the contact form at `https://www.overleaf.com/contact`.

3 State of the Art

There are already hundreds of smoking cessation apps available, but unfortunately, most apps are not scientifically validated [3]. In a study by Bold et al., only 11 apps had published results supporting their efficacy.

3.1 Feasibility of apps

How feasible is it to use apps? What makes an app effective? Maybe go into detail about how behavior change in general is hard (see Dries: only about 30% success rate, even for tobaccologists).

While many apps are available, the effectiveness of these apps may be limited [1], [5]. Luckily, it has been shown that digital interventions, through apps, can work [5], [6], when the right interventions are used. Even people who already make use of face-to-face care can find benefits in using digital help [5].

3.2 Description of existing apps

For some of the most salient apps: describe the methods they use. Are the methods scientifically based? Do they rely on scientific evidence?

From the literature, we see that a lot of apps do not use science-based interventions [4].

To collect information about the most popular smoking cessation apps, we searched the internet for the top-rated apps, and looked at Facebook groups to see what users recommend to

each other.

We looked at a few of the most popular smoking cessation apps, and we saw that they often employ these methods:

- **Progress trackers.** There are many variations of this; some track the number of days without smoking, some track the amount of money saved by not smoking, some track lifetime saved by not smoking.
- **Social aspects.** Some apps use a social aspect, like forums or other chat functions.
- **Planners.** Apps might help users plan designated smoke-free days, or plan a quit date.
- **Cognitive Behavioral Therapy (CBT)**. Some apps use CBT as a basis for their interventions.
- **Gamification.** Many apps use badges, mileposts, or levels to signify progress. Some apps also include games as a distraction for when a craving hits.

To make things more concrete, we will now look at a subset of the most popular apps, which we believe are scientifically based.

3.2.1 QuitSure

This app has published a study, showing its effectiveness [2], [4]. The study reports that 80.1% of the users maintained abstinence for at least 30 days after program completion [4].

The app is based techniques from psychology, CBT, and mindfulness [4].

In an onboarding process, the user is asked to think about their smoking quantity, patterns, past quitting attempts, and relapses. The app then assesses if the user is ready to quit. After onboarding, the user is intended to follow a program consisting of content and video exercises. This program requires 6 to 10 hours over 6 to 12 days to complete [4].

In their discussion, the researchers note that there is a high dropout rate of 59.4% [4], as a result of the high engagement requirement. Additionally, most relapses were a result of high cravings, for which the app did not have specific interventions after program completion.

Unfortunately, the study does not go into detail on what specific modules in the app contributed most to successful smoking cessation.

3.3 Behavior models

Found: HAPA, COM-B, TDF. Explain what each model does, on what timescale, and investigate how these models can be merged / used together.

Through our first meeting with Geert Crombez, we already came into contant with the Health Action Process Approach (HAPA) and Behavioral Change Techniques (BCT) models. In our

own search, we came across the Capability-Opportunity-Motivation-Behavior (COM-B) and Theoretical Domains Framework (TDF) models.

These models all serve a slightly different purpose and operate on different timescales.

The COM-B model [citation needed] describes the relationship between capability, opportunity, and motivation in influencing behavior. This means you can influence behavior by tweaking the other factors. The model can thus describe what factors you need to change.

HAPA [7], [8] makes a distinction between a preintentional phase and a volitional phase. In the first phase, a person develops their motivation and plan of action. Once this is "good intention" is sufficiently developed, the desired health action can be initiated. Now, in the volitional phase, this action needs to be maintained, which needs self-regulatory skills and strategies [7].

Compared to COM-B, HAPA focuses more on the psychological processes that drive behavior change, particularly the transition from intention to action and the maintenance of that action, rather than the physical or environmental factors.

We want to use a hybrid approach, combining models to find the best intervention for the user for each stage in the program. Further research is required into how the models interact, and determine the feasibility of this idea.

3.4 Related research topics

Related topics: how to use chatbot integration correctly? How to steer our interventions according to user feedback (either direct or indirect feedback)?

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A Appendix: Your first appendix

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B Appendix: Your second appendix

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